

in a smart sensor always comes from a semiconductor. The sensing principle consists of changing the transducer parameters to measurable electrical parameter and amplifying these to the point where digital electronics can start playing its role.

With miniaturisation in the process manufacturing, several optical or mechanical elements are becoming small enough where semiconductor manufacturing techniques can be applied directly. CMOS imaging sensors and MEMS sensors, which have gained a lot of popularity, are a few examples in this category.

MEMS have caught the imagination of many. What are your views on MEMS sensors? Which are the other types of smart sensors catching up?

The industry has progressed and sees a great potential in MEMS-based parts. Motion sensors are growing the most with applications such as mobile phones, gaming, remote controllers and automotive driving the rapid growth of these ICs. The gaming industry is making very smart applications based on accelerometers. The accelerometer is also useful to determine the shock and the applications can be designed such that the power supply to the critical sections can be cut before it impacts the real application.

MEMS magnetometer is an upcoming product that will drive the digital compass application for map navigation in mobile phones. Use of two-/three-axis gyroscopes is growing fast for user interface and image stabilisation in mobile phones, digital cameras and gaming.

Capacitive multi-touch sensors used in smart phones and tablet PC is another growing area.

How do you see the global and Indian markets for smart sensors? Which are the top five applications of smart sensors?

In India, smart sensors are still in a nascent stage as local manufacturing of mobile phones, gaming consoles and other handheld gadgets is yet to take off. But, these are the source of curiosity for many circuit designers, so the market should pick up in two-three years. The market is very small but should follow the world market trend as the production base is established here.

Globally, smart sensors—particularly, motion MEMS (accelerometers and gyroscopes), touchscreen and touch key controllers, and temperature sensors—are in huge demand.

Mobile phones, gaming, automotive, medical, LCD TVs and laptops are the most popular and innovative applications.

How is STMicroelectronics looking at smart sensor market?

Semiconductor companies, particularly STMicroelectronics, are very bullish about this market. The trend

is to add many value-added features in these products so that they can act smartly with minimal requirement of the MCU. This will reduce the overload of microcontroller software and allow easy integration into the circuit.

ST has introduced a smart sensor that combines a 3-axis accelerometer with a 3-axis magnetometer with all the signal conditioning and intelligence inside. It avoids the need of using two ICs and calibrating them so that they work in cohesion with each other.

Another major trend is to build an ecosystem around these sensors. Companies are developing drivers around major mobile platforms such as Android to allow for their easy integration with the rest of the platform. They are investing heavily on smart sensors to make the life easier, safer and trendy.

What kind of support is offered by STMicroelectronics to make designing easy using a smart sensor? Do you offer a ready-to-use platform where several sensors can be evaluated fast?

Smart sensors can be categorised based on analogue output and digital output. In analogue output, you need to use signal conditioning to get the desired form for final application. It requires a good knowledge of analogue circuit design. In digital output, the signal conditioning portion is generally integrated in the IC itself and the output comes on a serial bus. This needs good knowledge of microcontroller interfacing and programming.

There are, in fact, several platforms provided by STMicroelectronics. One of the latest platforms is ST's iNEMO (iNertial MOdule)—a sensor based module that integrates the inertial, magnetometer, gyroscope, temperature and pressure sensors in one module. It is a complete hardware platform for a variety of applications including virtual reality, human machine interfaces and robotics.

How do you see the future of smart sensors? What will be the new offerings from STMicroelectronics?

New generations of such smart sensors as magnetometers, microphones, pressure sensors and proximity sensors are expected to become popular in the future.

More and more intelligence in these devices will ensure easy interface, ability to work in adverse environments and better performance.

You can expect to have multi-sensor integration with an MCU in the form of a system-in-package in the future. ST's iNEMO board shows how it could look like, combining 9-10 degrees of freedom on one board. Such a system integrated in a package will have its own power supply and low-power RF for communication, paving the way for wireless multi-sensors for consumer electronics, home automation, health-care and much more. ●